Content

PRAT 1

Overview

General Intro of Our Search Engine

PRAT 2

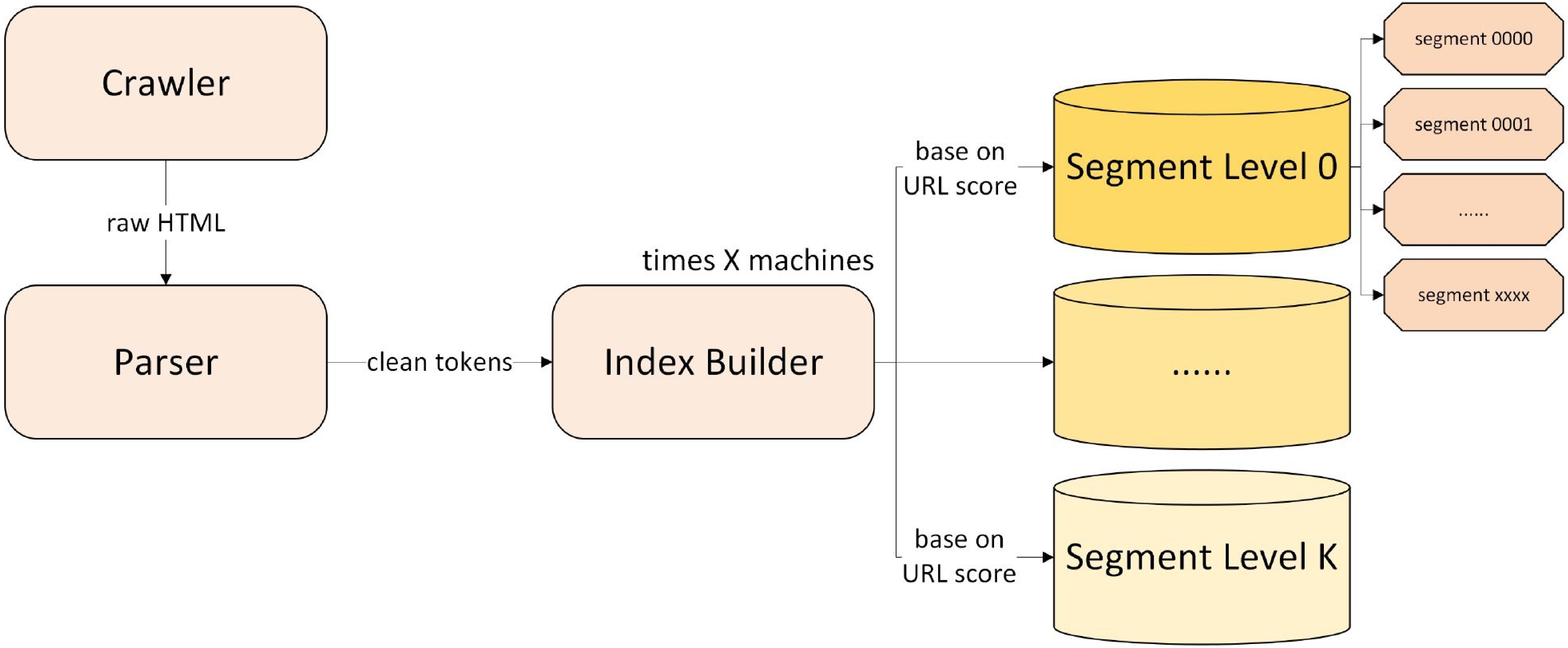
Introduce by Part

Detailed Introduction Part by Part

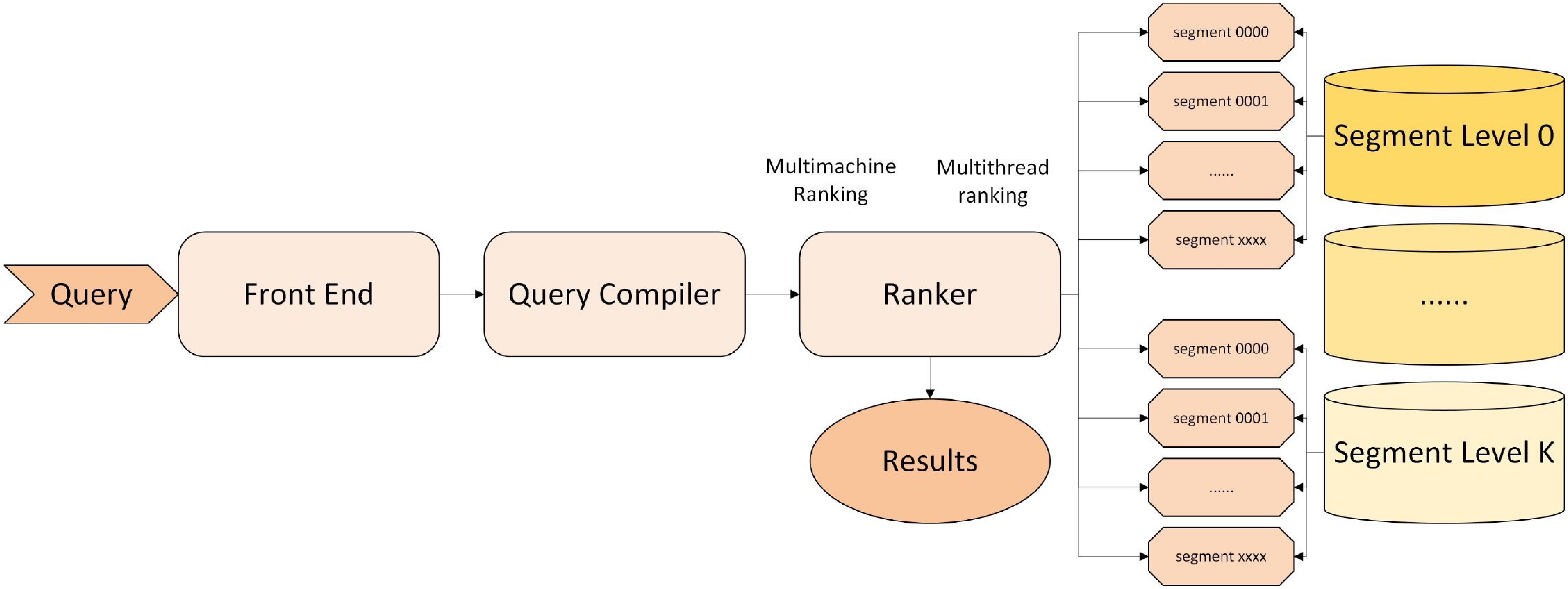
PRAT 3

Demo

Demo of the Result of Our Search Engine



Offline Index Construction



Online Query Serving

Line of Code

Parts	Line of Code
Common functions	~1900
Crawler	~4500
Parser	~1200
Index Builder	~5100
Query Compiler	~770
Ranker	~3100
Frontend	~500
Backend	~2400
Total	~21000

Section1

Offline Index Construction

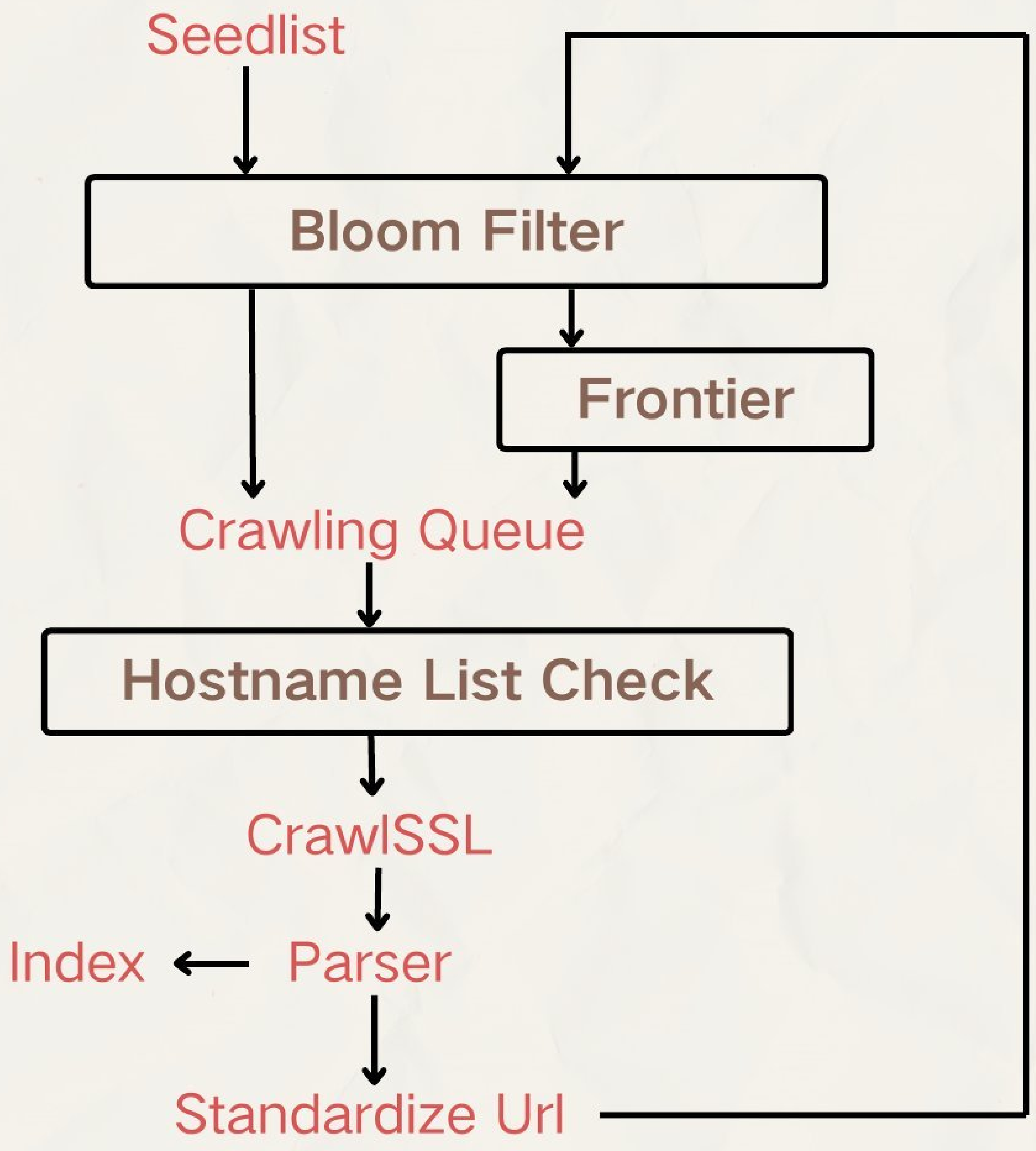
Components

each url

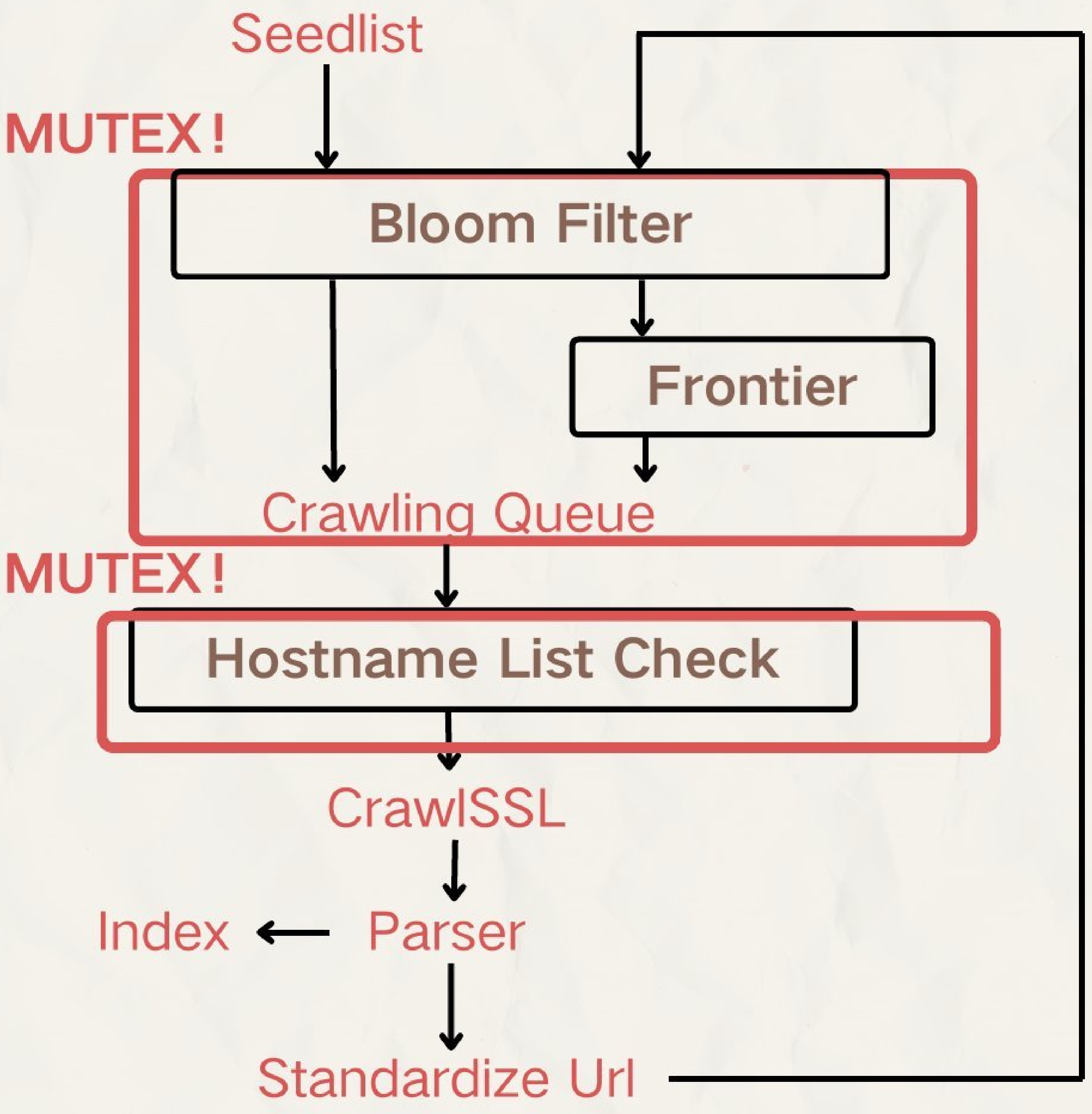
Bloom Filter	Deduplicate the urls
Frontier	Stores and organizes the global URL pool
Crawling Queue	Store candidate urls for next crawling
Hostname List Check	Check the robot.txt policy
CrawlSSL	Crawl the Url
SaveResults	Parse and save url, body, ...

Structure

single machine + single thread

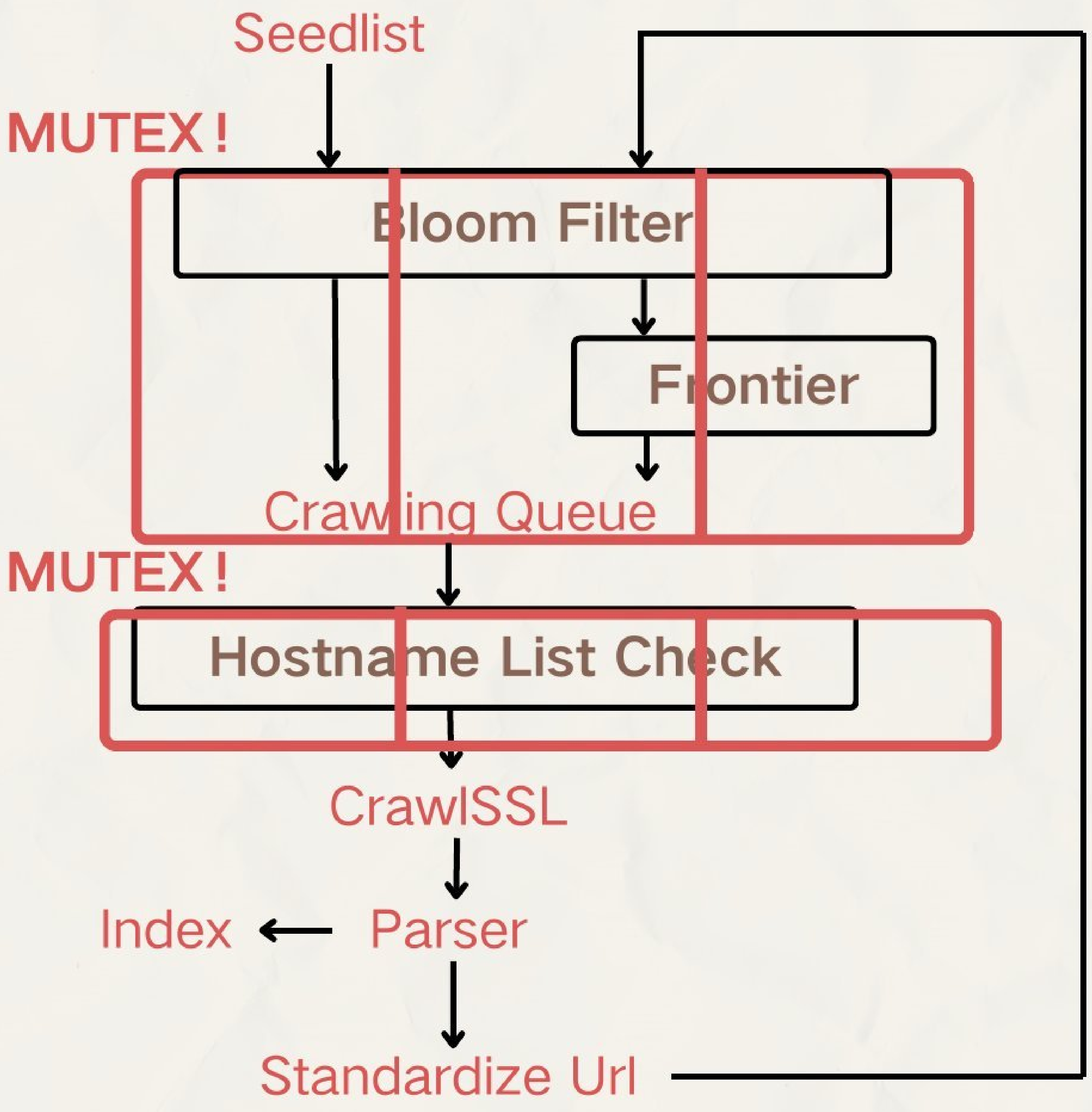


single machine + multi thread



Structure

single machine + multi thread



Problem:

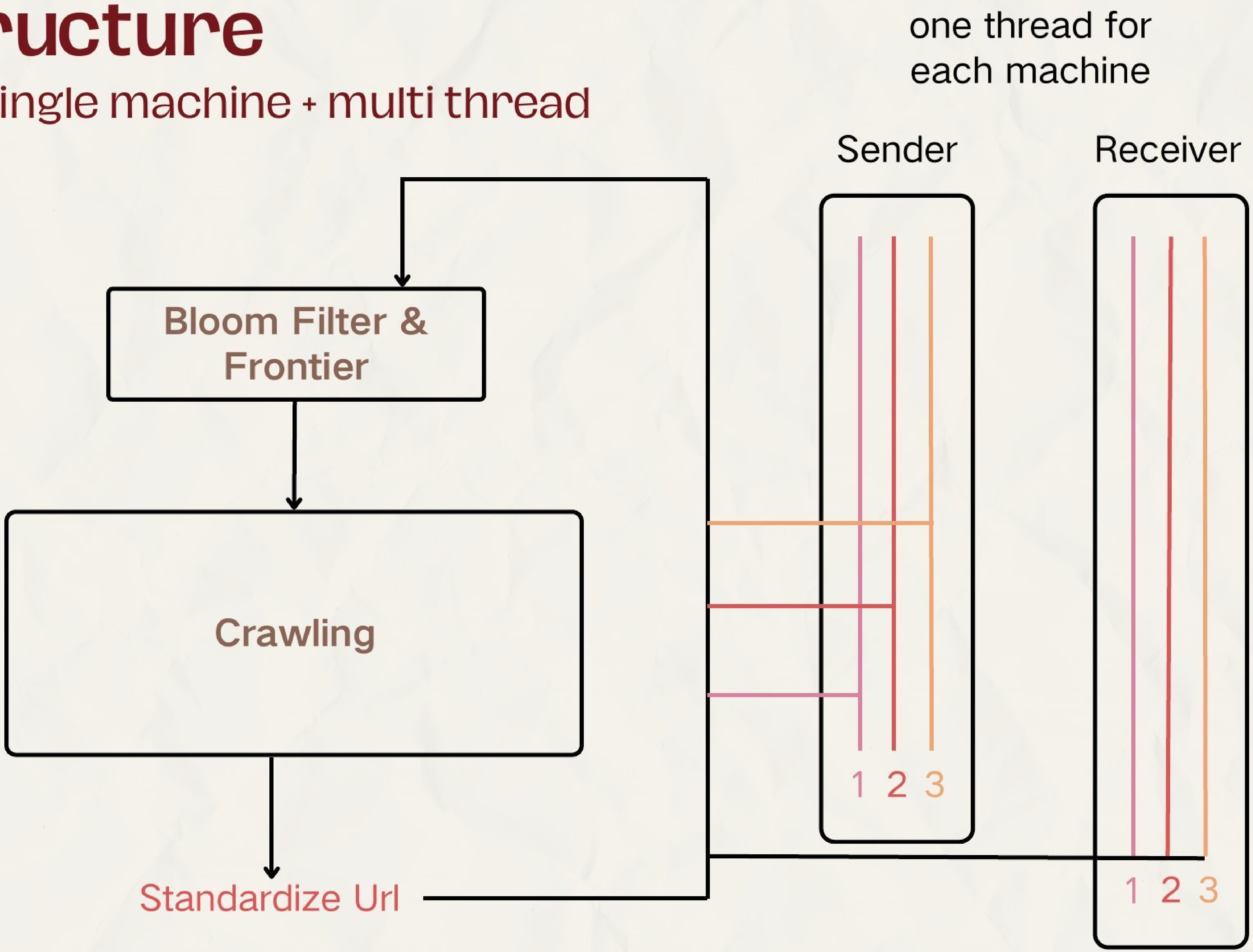
One big mutex -> The system not efficient when the worker thread++

Solution:

- Split the Bloom Filter, Frontier, and Hostname List into partitions
- Each partition has its own Bloom Filter, Frontier, and Hostname List
- Workers are assigned to specific partitions
- More workers can run simultaneously

Structure

single machine + multi thread



- **Hash** the url to put into the corresponding **buffer in sender**
- See if received something in receiver and **put back to frontier**

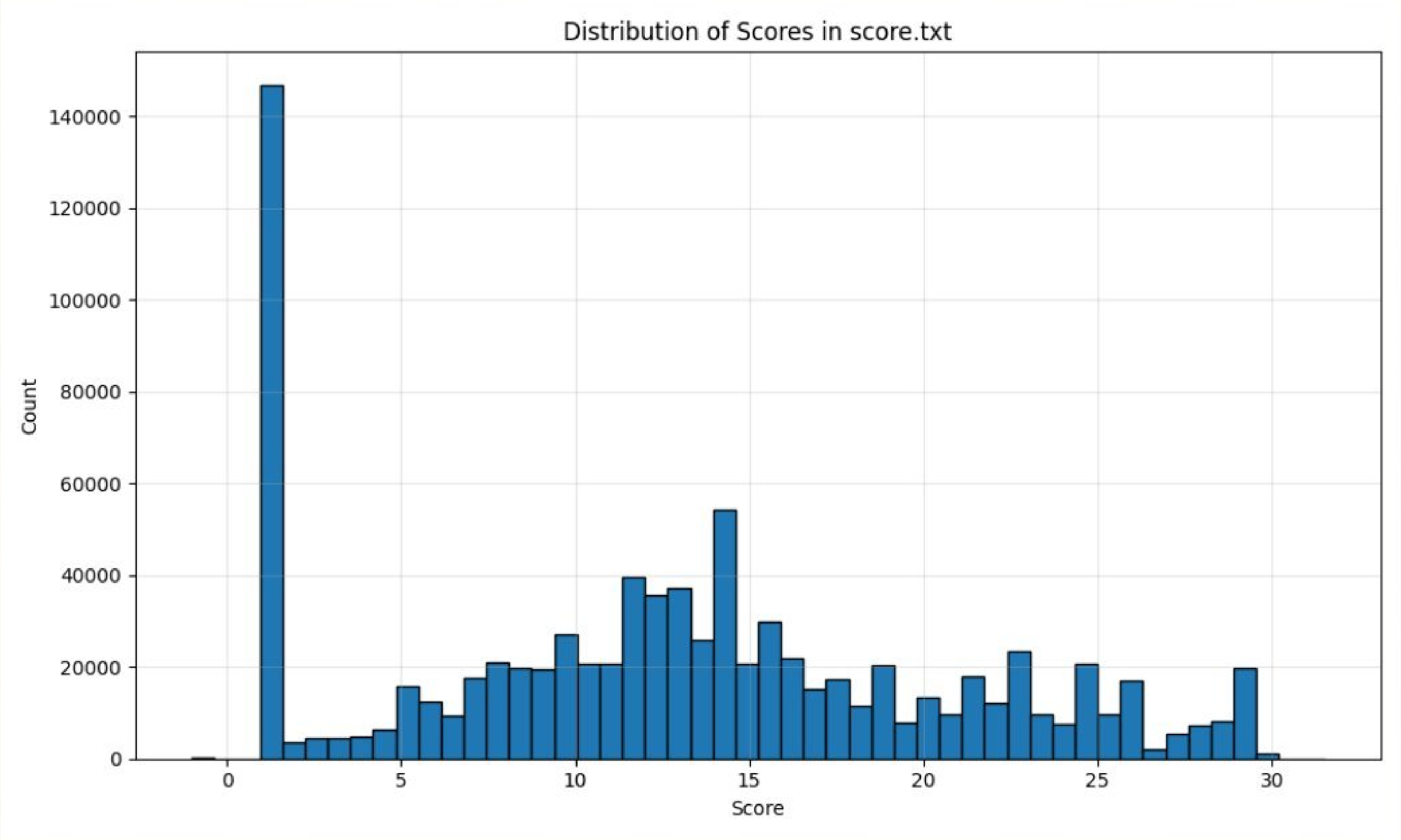
URL Standardize & Score

URL

- HTML entity decode
- http->https
- Search for redirect
  "u=" / "redirect="
- Delete ".txt/.pdf..."
- ......

Score

- Depth ⭐
- Https bonus
- Domain bonus
- Url length
- ......

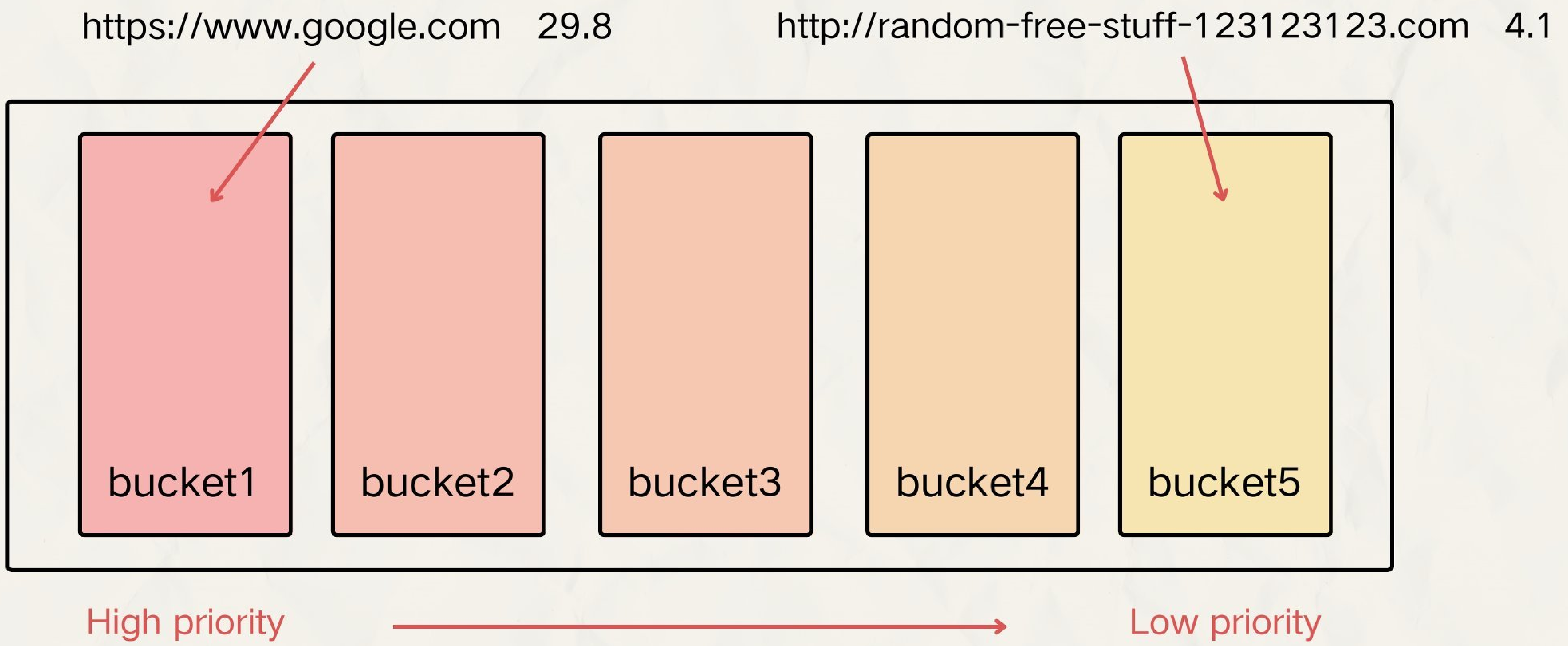


Prune low-score URLs early to reduce storage and improve frontier quality!

Frontier

Multi buckets with different priorities

URLs are assigned to buckets based on their scores.

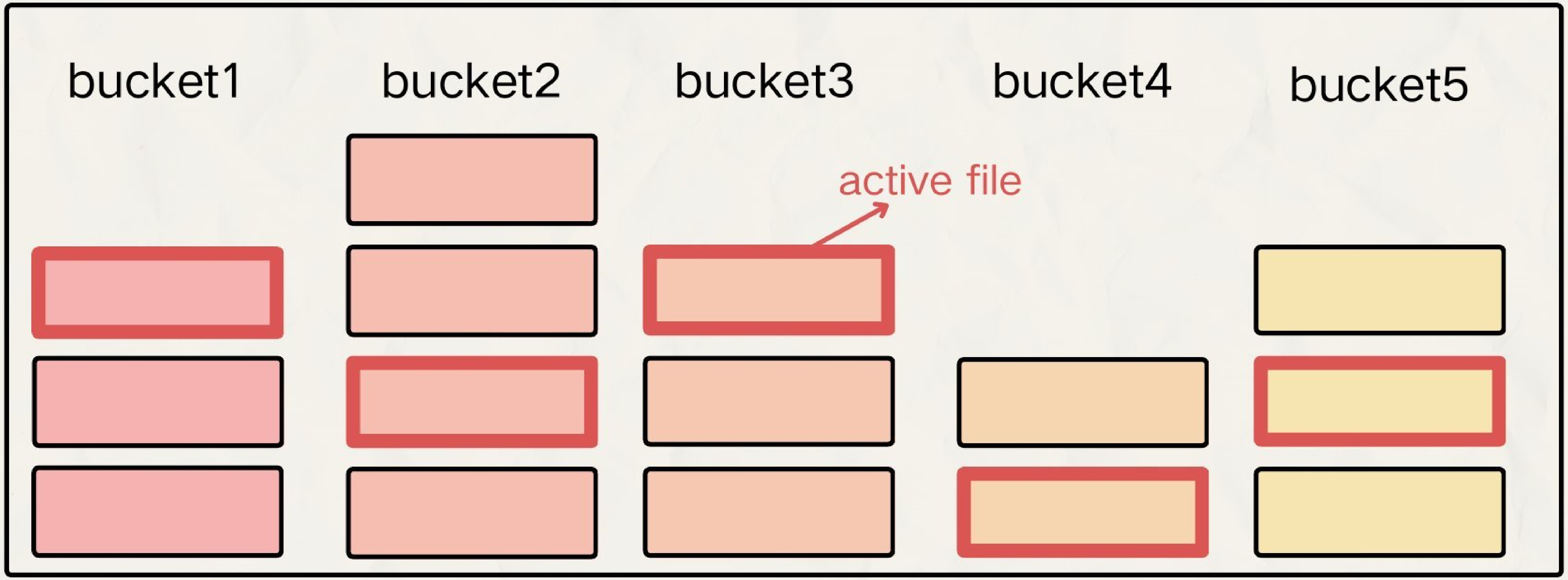


Frontier

Reduce memory usage and I/O cost!

Multi files per bucket, one active file at a time

Only active file is in memory.
All insertion and extraction is applied to this active file.



- Large file → split
- Small file → merge
- Too long time → rotates

↓

Only flush when active file changes !!

Hostname List

Store robots.txt and enforce crawl rules per hostname

Crawling Queue

URLs hostname

Hostname List Check

CrawlSSL

Politeness Control

robots.txt + crawl delay

Adaptive Backoff

Increase delay for failed hosts (e.g. 503)

Memory Bounded

Evict entries when full

Parser

HTML page

↓

Language Filtering

- <meta charset>
  Only UTF-8, ISO-8859-1, Windows-1252
- <html lang="...">
  Only English pages
- Non-ASCII Filtering
  Dynamic threshold to limit uncommon char

↓

Content Extraction

- Original title
- Cleaned title + body
- Links and base URL

↓

Text Cleaning

- Lowercases all alphabets
- Strips leading/trailing decorative chars
  @user → use, #tag → tag
  Keep special cases like C++
- Skips non-ASCII tokens
- Drops strange words
  >20 chars
  >4 special chars
  Pure numbers >5 digits
  Long (>10) alphanumeric ones

Index Building Process Design

- Word decoration
  - For term umich
    - **@umich** if in title
    - **%umich** if in url
    - **umich** if in body
- URL token normalization
  - Split URL and remove noise
    - https://umich.edu/student-life → umich, edu, student, life
  - Split CamelCase
    - https://ComputerScienceMajor → computer, science, major
  - Word breaking based on dictionary
    - https://machinelearningcourse → machine, learning, course

Index Building Process Design

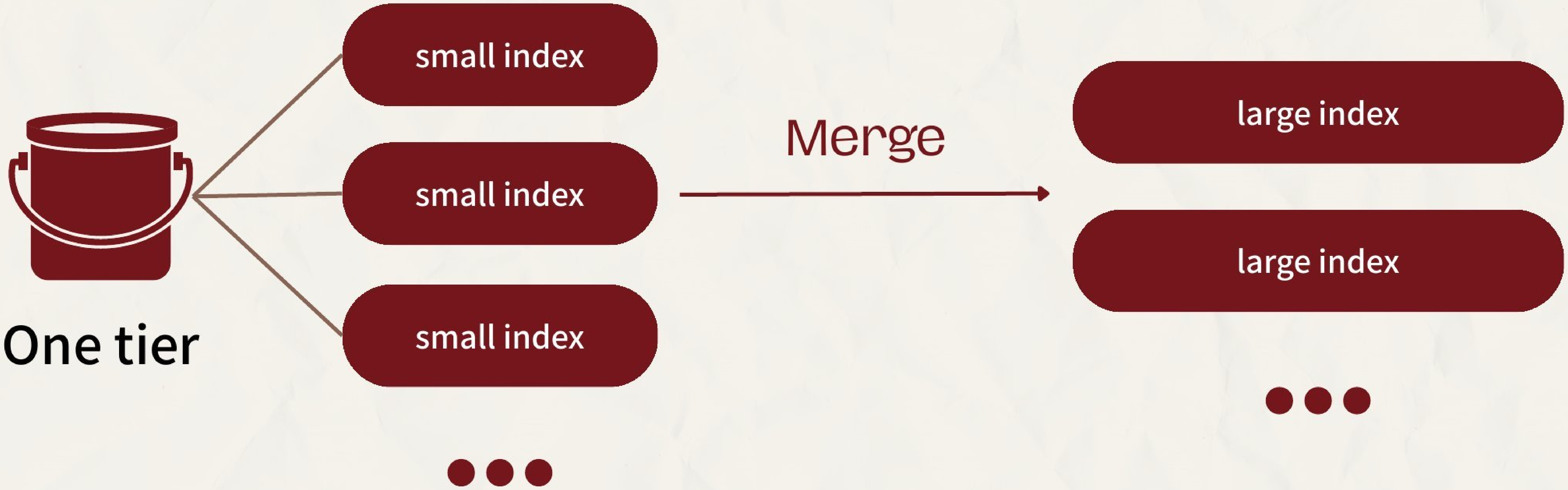
- Tier-based Index System
  - Pages are assigned to **different tiers** based on URL scores
  - Higher-tier index are prioritized during query time

Really Good!

	Tier0	Tier1	Tier2	Tier3	Tier4
static score:	>24	>22	>20	>15	>4
proportion:	7%	16%	28%	39%	10%

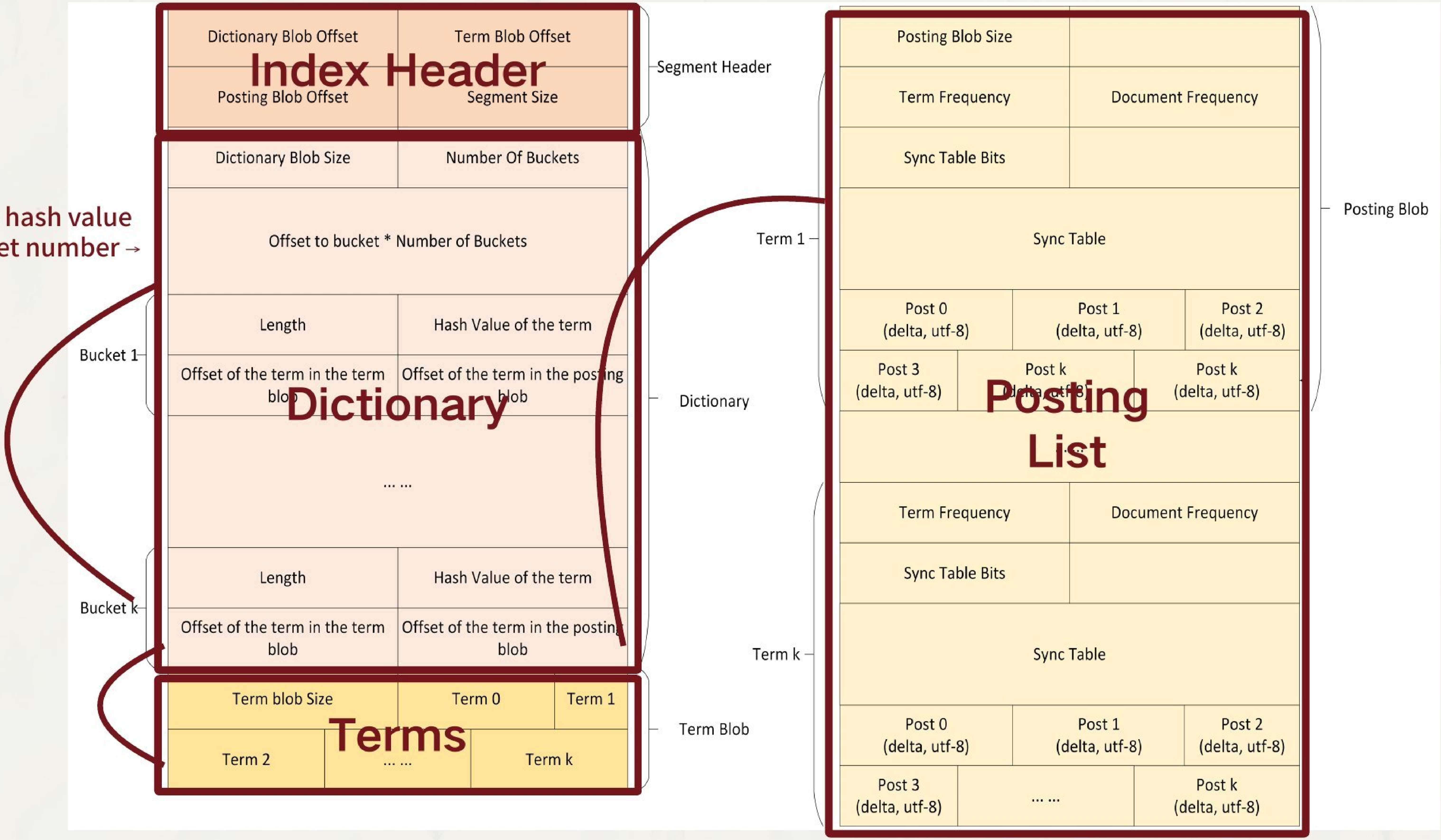
Index Building Process Design

- Checkpoint & Flush
  - When we gets 500 pages in hashtable, flush it to disk as a file
  - Can merge to bigger index (**Tune index size & count for better search speed**



How to find a Term's posting list

Term → hash value
→ bucket number →



Index Design decisions

target = 1001 0110

slot =
target >> (32 - N)=1

slot 1 →
2nd entry in Sync Table

SyncTable for SyncTable			
0	2	12	20
slot 0	slot 1	… … slot 2	

SyncTable		
entry #	absPos	offset in delta stream
0	10	0
1	90	2
2	150	6
… …		

Delta Stream					
10	+20	+70	+9	… …	+202

For size reduction

- delta encoding + sync table
- utf-8 encoding

Fixed checkpoint every 64 term occurance

Index Design decisions

Two-Level SyncTable Design:

Prefix-based approach

- Index based on absolute position prefix
- Problem:
  - ignores term frequency (tf)
  - buckets can be very uneven
  - long delta scans → slow speed

VS

Our SyncTable approach

- Insert checkpoint every 64 term occurrences
- Guarantees:
  - bounded scan (⩽ 64) in delta scan

Search speed x 64 !!!

ISR - to find a matching page fast

- **ISRWord** for single term
- **ISRDocEnd** for document end and doc data
- **ISRAnd，ISRPhrase，ISROr，ISRContainer**

DocEnd term design：

[docLen]
[url]
[title]
[depth]
...

OldDesign: [delta post] [docId]

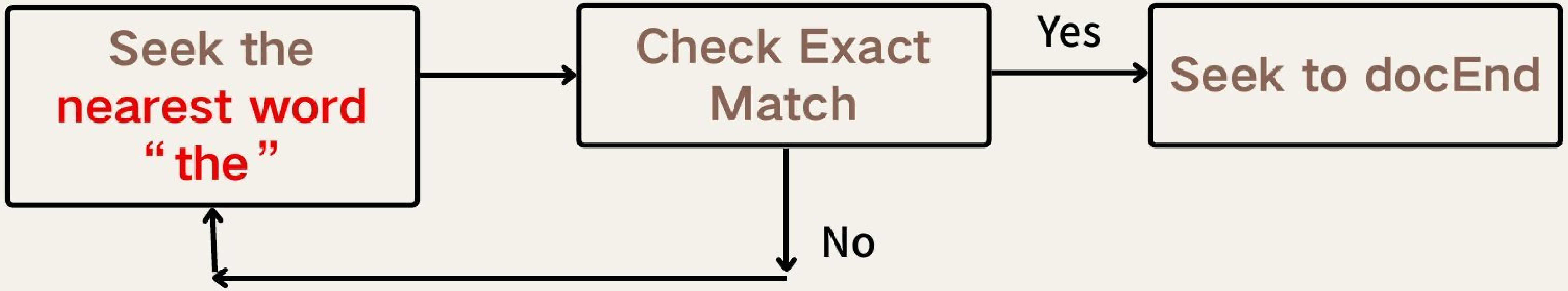
NewDesign: [delta post] [docLen] [urlLen] [url] [titleLen] [title] [depth]

avoid extra disk IO

ISR - Phrase search: “the quokka”

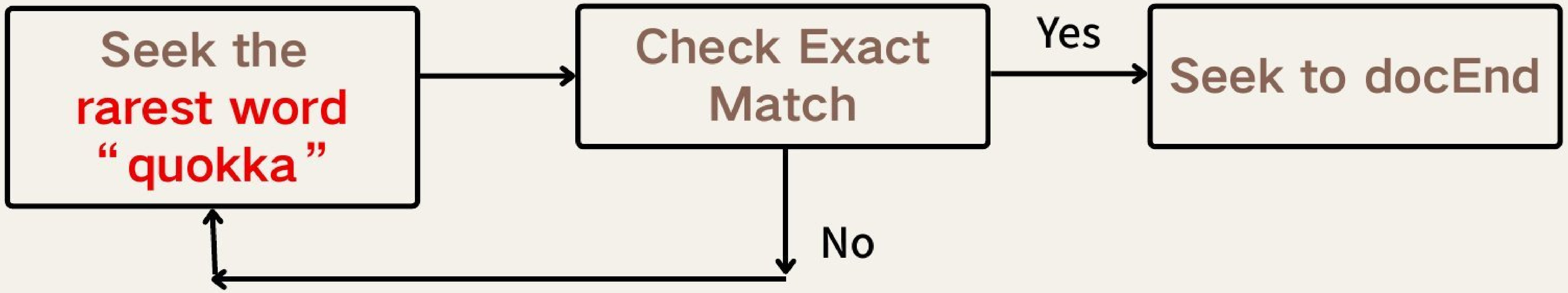
Slow

Too many candidates



10x faster

Few candidates



Offline Deployment Result

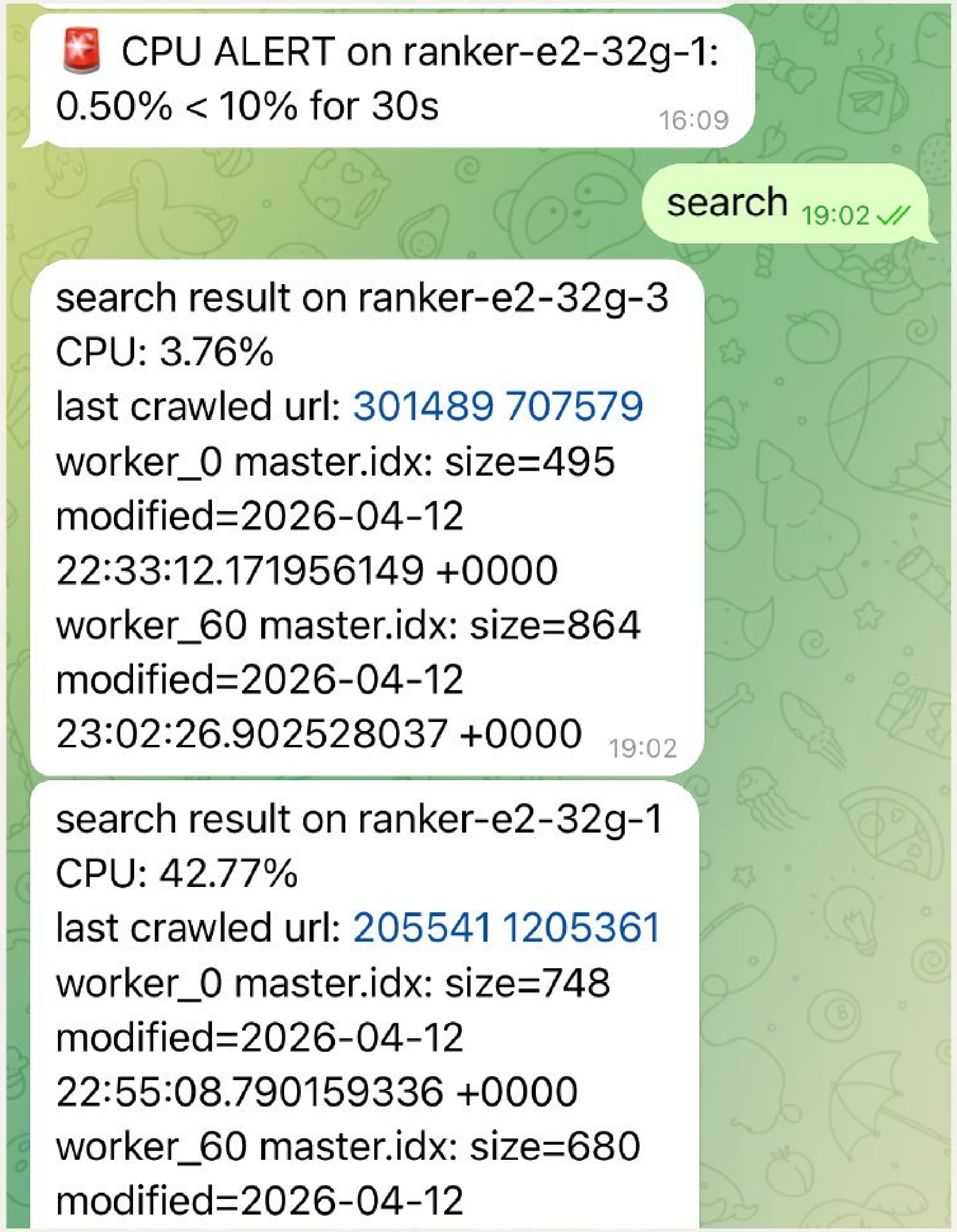
- 5 Virtual Machine with 8 vCPU, 32G Memory(e2-standard-8)
- 30 million Index size (Still Crawling lol
- Query language: English

System Capabilities

- Easy continue from crash
- One failure ≠ system failure
- Real-time monitoring from anywhere

Fantastic Monitoring Tools！！

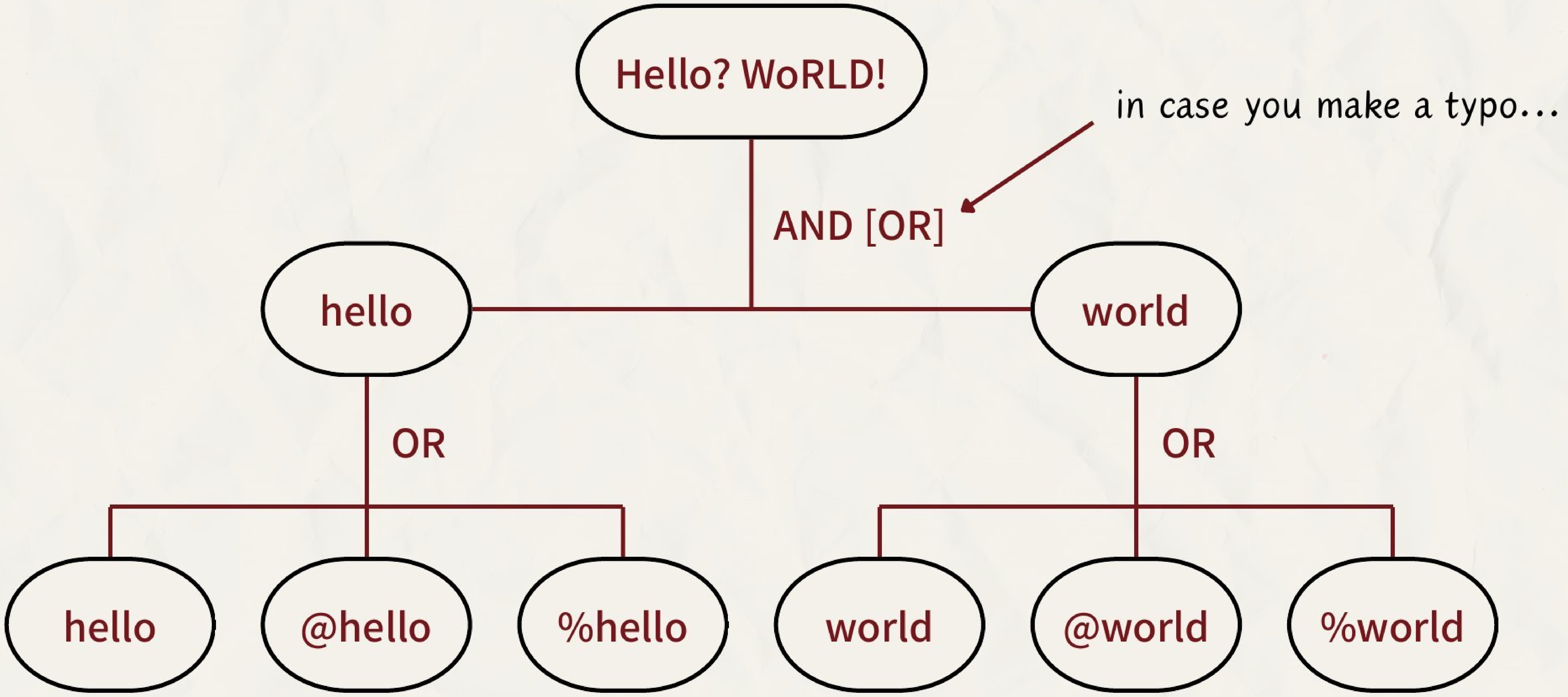
Search bot



Section2

Online Query Serving

From Query to ISR Tree



Syntax List

“phrase words”	Exact consecutive phrase
a b, a AND b	Both must appear in same doc
a | b, a OR b	Exact consecutive phrase
NOT a, -a	Exclude term from results
(expr)	Exact consecutive phrase

Ranking Heuristics - Static score

Yes:

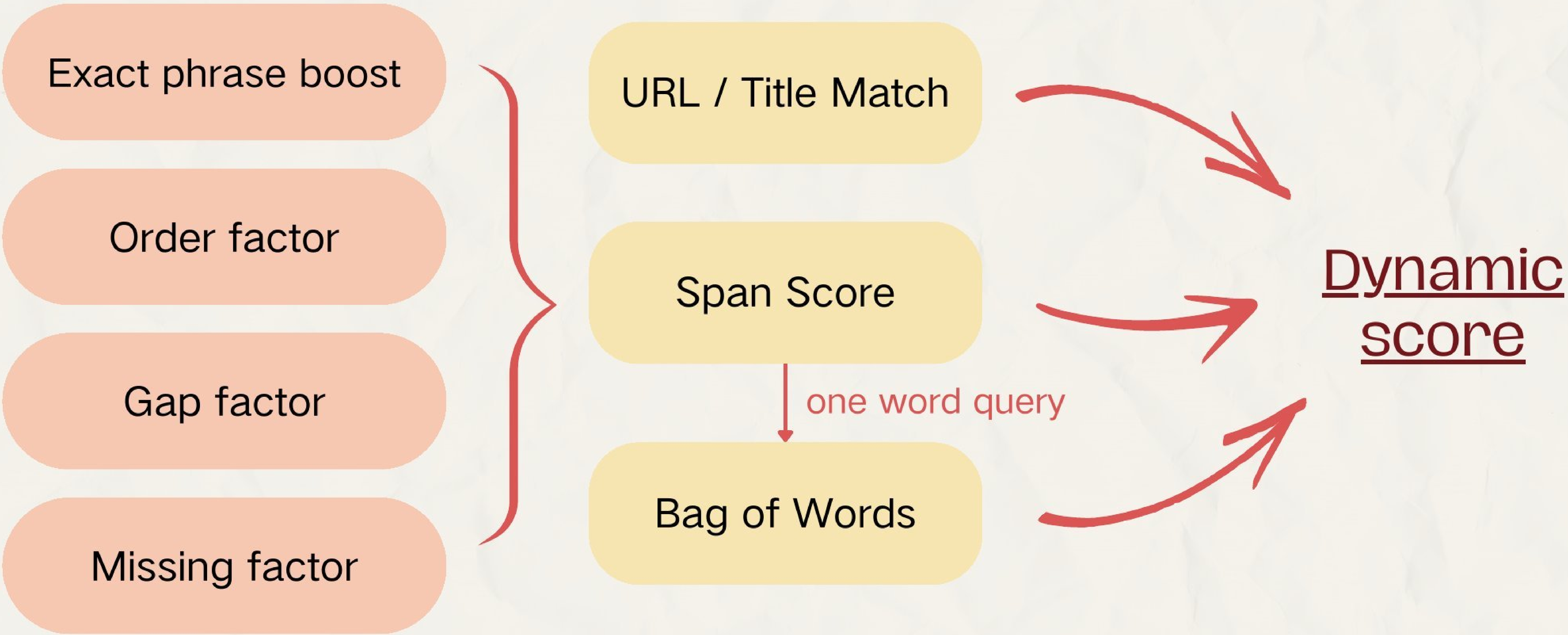
- HTTPS
- root path
- .gov/.edu/umich.edu
- Wikipedia
- www. subdomain

No:

- Large URL length,
- Far from seedlist
- Misc numbers in URL
- Long titles
  - Title > 200 chars are discarded

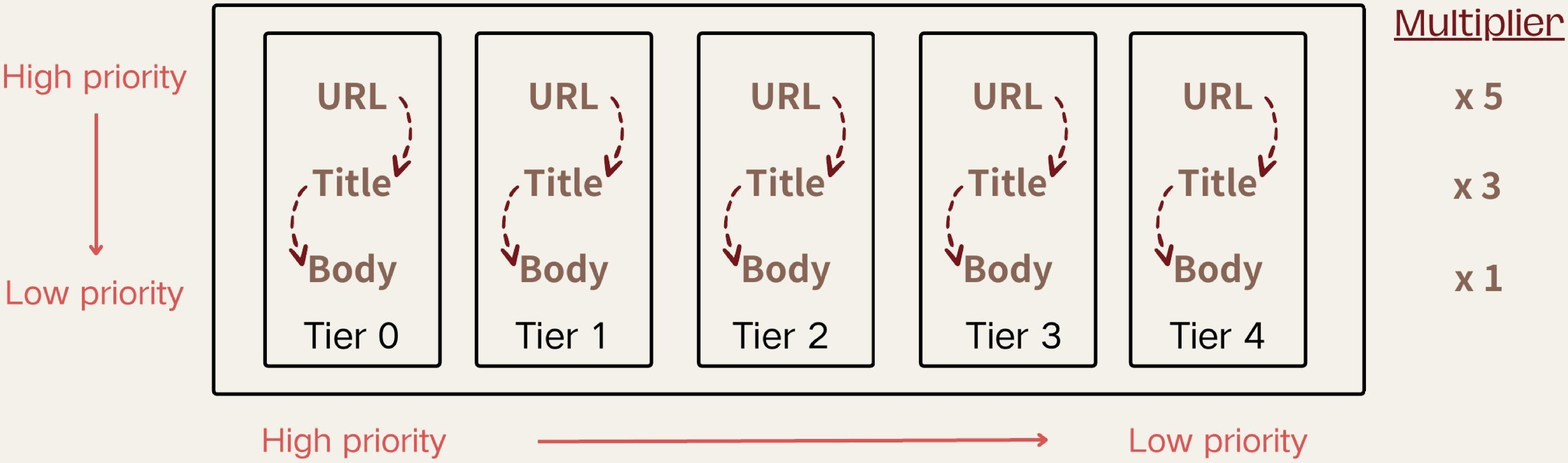
Final score is a weighted sum of static and dynamic score

Ranking Heuristics - Dynamic score



Ranking Strategy

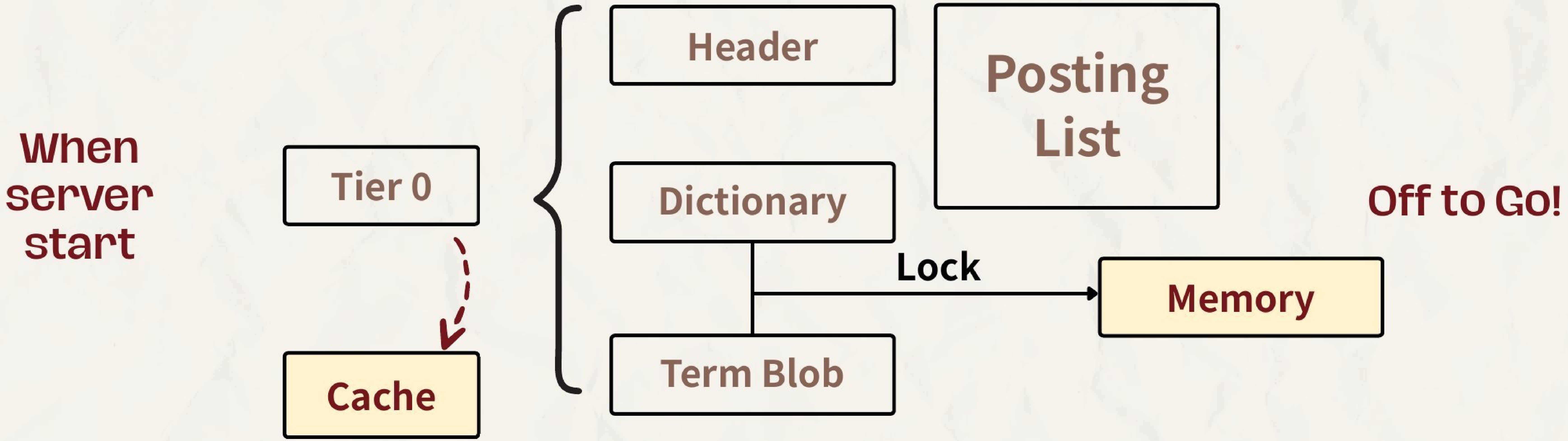
Tier-first, Three-pass Ranking



Ranking Strategy

Cache warming

- At the start of server, mmap the indexes into memory, set MAP_POPULATE flag for index of Tier 0 to trigger page fault for cache warming
- Lock the TermBlob and DictBlob of Tier 0 index in memory (most frequently used!)



Ranking Strategy

Domain Capping

- Discount results in same domain (0.65× per additional result)
- Cap each domain for at most 3 results

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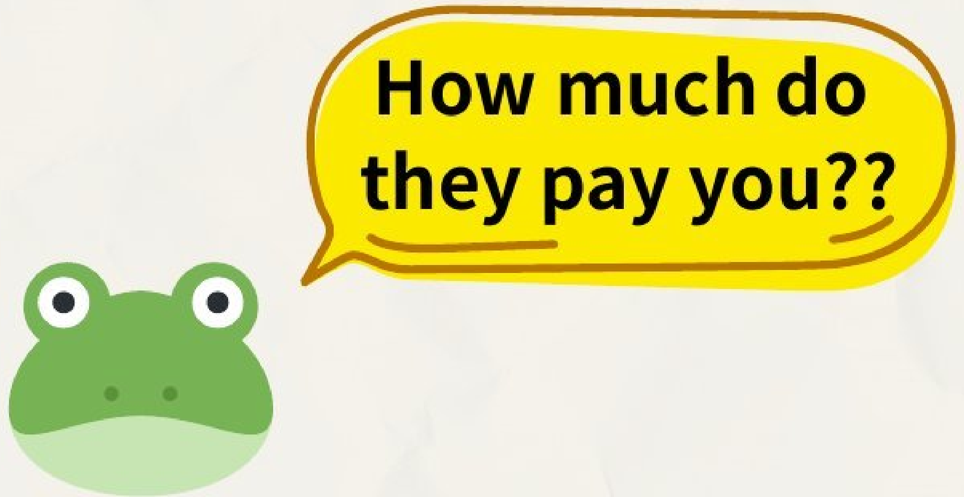

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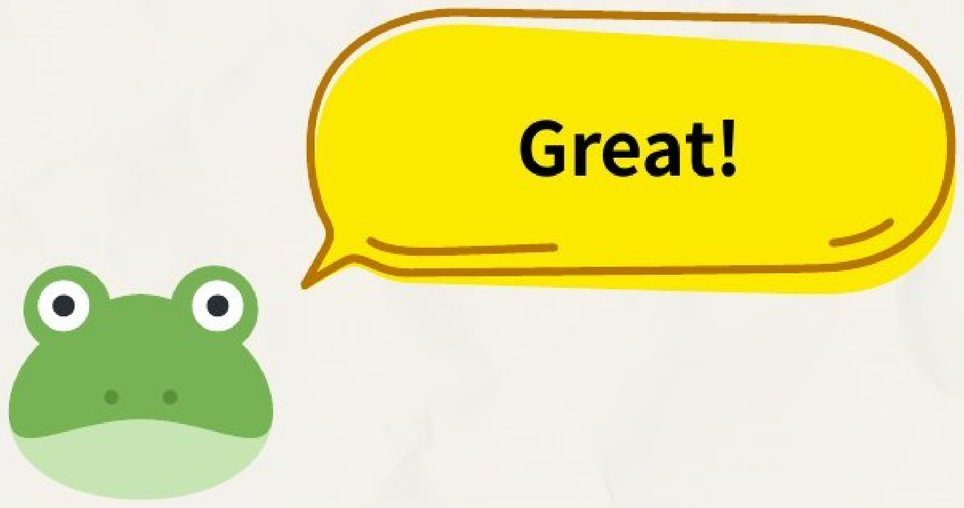

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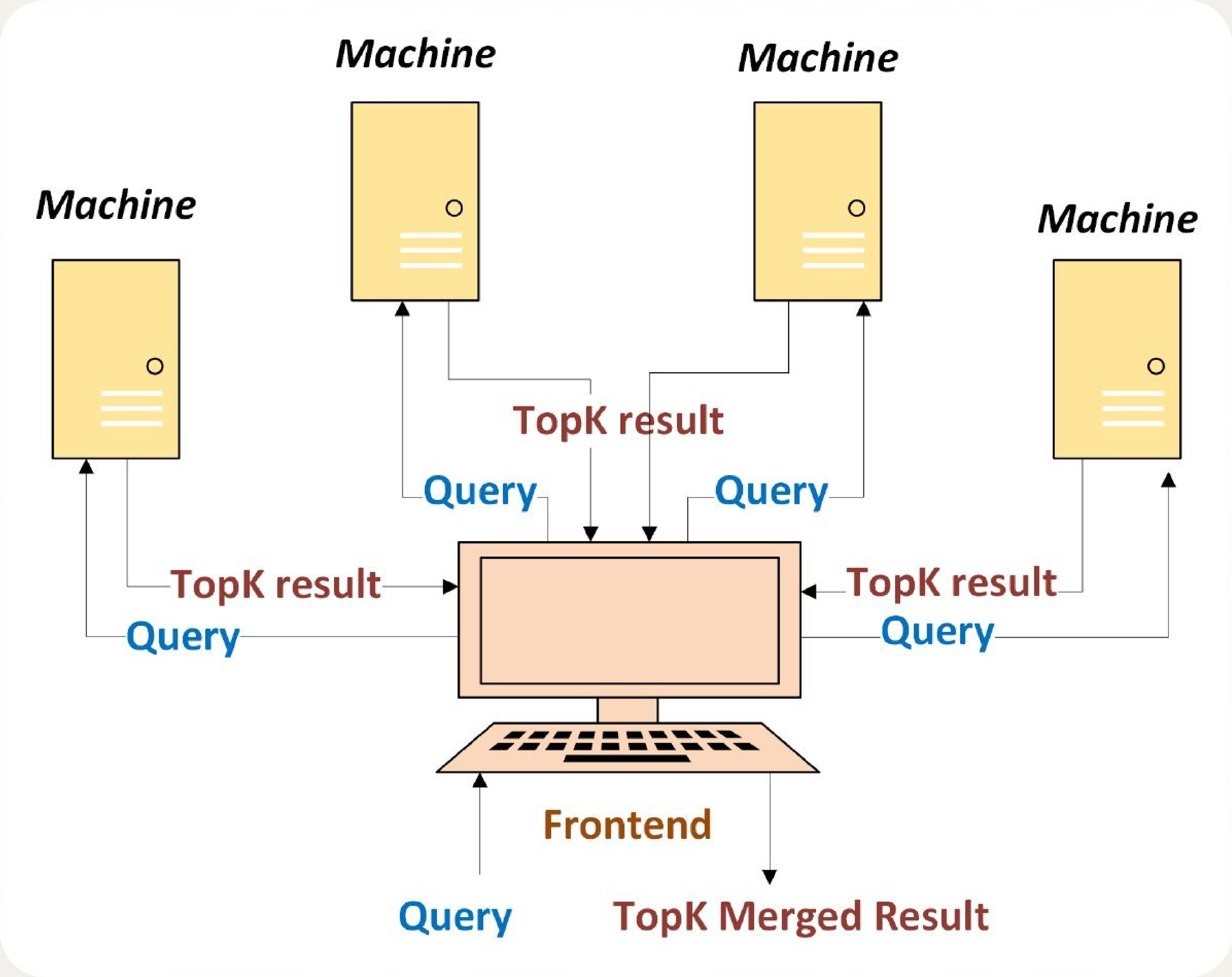

Multi Machine Ranking

The Ranking Hat

Demo

The Ranking Hat
(Your personal internet detective 🤨

amazon -basin

found 10 results about "amazon -basin" in **0.968 seconds** | **945838** docs · 1018512 ranked/sec

https://www.amazon.com

Amazon.com

Expand snippet

https://en.wikipedia.org/wiki/Amazon.co.uk

Amazon (company) - Wikipedia

Expand snippet

https://en.wikipedia.org/wiki/Martinique_amazon

Martinique amazon - Wikipedia

Expand snippet

https://en.wikipedia.org/wiki/Amazon_Prime_Video

Amazon Prime Video - Wikipedia

Expand snippet

https://www.amazon.co.uk

Amazon.co.uk

Expand snippet

The Ranking Hat
(Your personal Internet detective 😉

archive of our own

found 10 results about "archive of our own" in **0.943 seconds** | **430218** docs · 567319 ranked/sec

https://archiveofourown.org/

Home | Archive of Our Own

Collapse snippet

turning it on! Work Search tip: "sherlock (tv)" m/m NOT "sherlock holmes/john watson" more than 77,560 fandoms | 10,570,000 users | 17,340,000 works The **Archive of Our Own** (AO3) is a project **of** the Organization for Transformative Works. You can join by getting an invitati

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https://archiveofourown.org/works

Latest Works | Archive of Our Own

Expand snippet

---

https://archiveofourown.org/diversity

Diversity Home | Archive of Our Own

Expand snippet

---

https://www.wikihow.com/Post-a-Fanfiction-to-Ao3-(Archive-of-our-Own)

How to Post a Fanfiction to Ao3 (Archive of our Own)

Expand snippet

The Ranking Hat
(Your personal Internet detective 😉

chiles v. salazar

found 10 results about "chiles v. salazar" in **0.461 seconds** | **859** docs · 2031 ranked/sec

https://en.wikipedia.org/wiki/Chiles_v._Salazar
Chiles v. Salazar - Wikipedia
Expand snippet

https://www.advocatesforyouth.org/statement-on-chiles-v-salazar/
Statement on Chiles v. Salazar - Advocates for Youth
Expand snippet

https://www.scotusblog.com/cases/case-files/chiles-v-salazar/
Chiles v. Salazar (Conversion Therapy) - SCOTUSblog
Expand snippet

https://outnebraska.org/chiles-v-salazar-pending-scotus/
Chiles v. Salazar – Pending Supreme Court Case – OutNebraska
Expand snippet

https://www.law.cornell.edu/supremecourt/text/24-539
CHILES v. SALAZAR | Supreme Court | US Law | LII / Legal Information Institute
Expand snippet

THANKS

The Ranking Hat